Collaborative, Open-Source Projects as Innovative Work-Based Learning Opportunities for Underserved Computer Science Students

Abstract: *Internships and work-based learning experiences help college students secure tech jobs but often exclude adult learners, low-income students, and other underserved groups. In response, Green River College, Mentors in Tech, and CodeDay developed a work-based learning model where computer science students collaborate on open-source software projects with guidance from industry mentors. Students reported gains in communication, teamwork, project management, and technical skills, and felt better prepared for employment. The model was especially valuable for adult learners and those unable to access traditional internships. Faculty also noted that open-source projects effectively simulate workplace experiences and enhance real-world readiness.*

Objective/Purpose

Internships during computer science (CS) programs are vital in helping college students secure high paying jobs in the tech industry upon graduation (Strada, 2024). However, during and after COVID-19, students faced a diminished job market: internship opportunities dropped nearly 50 percent (Stansell, 2020a), and job openings for graduates fell over 60 percent from pre-pandemic levels (Stansell, 2020b). These trends compounded longstanding inequities in internship access, particularly for adult learners, low-income students, and students of color (Francisco, 2020; Singer, 2023; Strada, 2023).

At Green River College, a public community college in Washington's Puget Sound—an economically vibrant region anchored by the tech industry—these issues were especially urgent. In response, faculty collaborated with Mentors in Tech (MinT) and CodeDay to develop a novel work-based learning model: Building Open-Source Opportunities for Students in Tech (BOOST). BOOST was designed to integrate internship-equivalent experiences directly into the curriculum of the Bachelor of Applied Science in Software Development program. This proposal explores how BOOST emerged in response to institutional constraints and student needs, and how it repositions faculty roles, curriculum, and mentorship within community college contexts.

About the Organizations

Green River College (GRC) is a federally designated minority-serving institution (MUREP, 2023) that offers two- and four-year technology degrees. Its Information Technology department received national recognition for equity in STEM education (LeRoi-Schmidt, 2022). Green River is one of 187 U.S. community colleges authorized to confer bachelor's degrees (CCBA et al., 2024).

CodeDay is a nonprofit that creates open-source internship programs connecting students with tech professionals as mentors. Mentors in Tech (MinT) is a nonprofit focused on helping students at accessible, affordable institutions navigate and launch careers in technology through structured mentorship (Mentors in Tech, n.d.).

About the Model

MinT and CodeDay worked with GRC faculty to recruit industry mentors and identify relevant open-source software (OSS) projects. This collaboration circumvented resource barriers that often prevent community colleges from building traditional internship pipelines.

The BOOST model embeds a comprehensive work-based learning model into the academic curriculum, with three key components:

1. Aligned classroom and industry-relevant learning
2. Application of academic, technical, and employability skills in collaborative projects
3. Structured mentorship by active professionals in the tech industry

BOOST is implemented as a 10-credit, two-course capstone sequence in GRC's applied bachelor's program. Students work in teams on OSS projects, guided weekly by mentors and supported in-class by faculty. Each student completes at least six hours of in-class work and four hours of out-of-class collaboration. Paid stipends support participation among students with financial or work obligations, addressing common barriers to unpaid internships (Menezes et al., 2022). This structure reflects an intentional faculty strategy to create more inclusive and workplace-relevant learning opportunities within the bounds of an academic program.

Theoretical Framework

Experiential Learning Theory (Kolb, 1984) is foundational to the integration of internships in higher education. In this theory there are four stages of career development beginning with concrete experiences such as internships, ideally coupled with the second stage, an opportunity for reflective observation and the third phase, abstract conceptualization such as linking the internship to classroom theory. Finally, the student returns with new strategies for projects, termed active experimentation. In the OSS model capstone internships provide students with experience working with various software programs in diverse settings with an opportunity for weekly structured reflection and classroom time. This framework underpins the integration of project-based OSS work, reflective practices during class time, and iterative learning cycles that align with real-world software development. Faculty used this theory to design a structured yet flexible curriculum that responds to the complexities of adult learners' needs and simulates industry dynamics.

Additionally, Betz and Hackett's (1981) career decision self-efficacy theory informed the program's design, particularly through:

- Performance accomplishments: Contributions to OSS projects visible to industry

stakeholders
- Vicarious learning: Mentors modeled problem-solving approaches, increasing students' sense of agency

These frameworks provided the conceptual grounding for faculty to align mentoring, coursework, and professional learning in a cohesive curricular design responsive to diverse learners.

Literature Review

Internship experience plays a critical role in the tech workforce, with work and internship experience being some of the most cited considerations when hiring a tech professional (Stepanova et al., 2021). A report from Strada Education Foundation found that students who completed an internship were "more confident in their ability to communicate their skills and experiences to potential employers than those who have not participated in internships" (2023).

However, many internships and other work-based learning opportunities are rife with inequity (Francisco, 2020). Research (Strada, 2023) has found that "70 percent of first-year students expect to have internships, yet fewer than half of seniors report that they have had one" (para. 7). Part of this asymmetry is that internships are often unpaid, making it difficult or impossible for lower income learners, working learners, or adult learners with caretaking obligations to participate (Francisco, 2020). Students of color, particularly Black and Latine students, are also "significantly less likely" to participate in internship opportunities (Strada, 2023).

Further, students applying to internships who attended "lesser-known public universities said they felt at a disadvantage compared with their peers" who attended more well-known institutions (Singer, 2023). In addition, some research (Van Noy & Jacobs, 2012) indicates that employers have more negative perceptions of community college graduates than their university counterparts. These perceptions add barriers for community college students when they enter the job market, but community college baccalaureate research (Soler & Bragg, 2015) has shown that work-based learning, such as internships, can improve employers' perceptions of community college students.

Barriers are not only structural but also psychological. In a qualitative study of 533 undergrads, low self-efficacy was the largest identified theme among students who did not get an internship (Kapoor & Gardner-McCune, 2020). These findings point to the need for curricular models that provide real-world preparation while intentionally cultivating confidence and belonging, particularly as a STEM professional—roles faculty are increasingly being asked to take on (Killpack, 2025; Newman et.al, 2024).

Modes of Inquiry

To study the BOOST model, we used a developmental evaluation approach. Developmental evaluation “supports development of innovation and adaptation in dynamic environments” (Patton 2011). Developmental evaluation is appropriate for work (a) that is innovative in design and (b) that tackles complex circumstances (Patton, 2011). This project employed an innovative approach to work-based learning in the tech industry through meaningful mentorship and open source projects, satisfying part (a), and was born in response to the COVID crisis and rampant inequities in the field, certainly complex circumstances (b). In such conditions, developmental evaluation can be beneficial in testing and improving iterations and surfacing issues along the course of the project (Patton, 2011).

The evaluation addressed three guiding questions:

1. In what ways has the BOOST model supported student learning and job readiness?
2. How, if at all, has the model advanced equity in access to work-based learning? 3. What lessons can inform future iterations of the program?

This approach supported a collaborative faculty role in evidence-informed curricular refinement. **Data Sources**

The data corpus included:

- Qualitative interviews with students, faculty, and program staff
- Program artifacts, including demographic surveys, course syllabi, and project descriptions

Two cohorts of students who participated in the BOOST model were interviewed toward the end of their respective terms. Students were interviewed in five groups, based on their team projects, with 18 total students participating. Of the 18 students, 9 were aged 25 or older, and seven students were career changers. Three of the interviewees were military veterans, and about one quarter of the group identified as female.

A MinT employee and two faculty members at GRC, one who designed the program and administers the project and another faculty member who is the lead course instructor, were also interviewed. For all interview sessions, a semi-structured interview protocol was used.

Interviews were recorded and transcribed using a transcription service. Using an open-coding structure, the interviews were coded and analyzed using a computer assisted qualitative data analysis software, MAXQDA. From these codes, we began sorting and aggregating codes into major themes. We engaged in an iterative process while remaining close to the data. Overall, we followed Robert Yin’s (2016) approach for disassembling data (i.e., creating open codes), reassembling data (i.e., categorical coding and looking for intersections of open codes), and interpreting data (i.e., drawing out themes from the coded data).

Findings

We found that the students found the project relevant, particularly in terms of communication, teamwork, project management, and developing computer language proficiency. They discussed the learning as occurring in a triumvirate – that is learning from (a) other students in their group, (b) the mentor, and (c) the course instructor. All students shared they felt better prepared for a job having participated in the BOOST model and more prepared to tackle the job market after the capstone project course. This mode of internship was also helpful to adult students and those who may not have been able to complete a traditional internship otherwise.

While reflecting on the capstone class, students felt more prepared to learn new technologies, gain real-life experience for interviews, and build confidence in their technical skills and ability to perform well in professional roles. They also discussed the surprise of realizing that software engineers have different perspectives on how things should be done and the importance of compromise and considering different solutions.

The students cited their mentors as being incredibly helpful, opening doors on a technical and professional level. One student described getting feedback from their mentor, a Google employee, as "mind-blowing" and shared that the mentor had invited them to a Google forum. Mentors were found to provide insightful feedback and support, especially in new software programs that the students were unfamiliar with.

Faculty members highlighted that focusing the work of the capstone on open-source projects, rather than working directly with an employer, was a significant benefit to the BOOST model. They shared that the experience gained from open-source projects is more relevant to future work, carries more weight on resumes and with prospective employers, and has more of an impact on the field than traditional employer-based internships.

Students cited the financial compensation for the capstone project as helpful but unexpected; many students indicated it was not a deciding factor for participating in the program, although it did help encourage more robust efforts in the course. Further findings relating to the project's contributions to equitable outcomes for underserved populations are forthcoming, including findings related to graduation and employment.

Scholarly Significance

The BOOST model demonstrates how faculty can respond to institutional constraints and structural inequities through innovative curriculum design. By embedding OSS mentorship into credit-bearing coursework, GRC faculty reimagined work-based learning to center inclusion, relevance, and mentorship—core concerns in contemporary postsecondary teaching.

BOOST expands the traditional faculty role to include partnership-building, mentorship coordination, and co-design with industry—all while sustaining academic rigor. It also provides a model for how community colleges can leverage their applied degrees to offer equitable, career connected learning experiences without relying on competitive internship pipelines.

Findings from this evaluation contribute to research on curricular innovations in STEM education, particularly within under-resourced institutions serving diverse learners. They also raise important questions about the scalability of mentor-supported, project-based learning in community college baccalaureate programs. Future research may explore longitudinal outcomes such as job placement and persistence in the tech workforce.

To evaluate its merit for replication on a broader scale, it must be tested at multiple colleges with additional mentors and open-source projects. We envision that this pilot of the BOOST model and the resulting evaluation can be used by other CS programs as they design innovative, equitable work-based learning opportunities for students to help them more competitively prepare for the workforce upon graduation.

This study aligns with Division J Section 4's focus on faculty innovation, curricular design, and equity-minded pedagogy in a changing postsecondary landscape. It underscores how faculty, when supported to experiment and collaborate, can create new pathways to opportunity for students historically excluded from high-value work-based learning.

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